

DBDOME Installation Guide

Installation, verification, upgrade and rollback

Product	DBDOME Database Security and Monitoring Platform
Applies to	DBDOME 2.02.x
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Audience	System administrators and DBAs performing installation
Classification	Customer deliverable

DBDOME Installation Guide

1. Scope

This guide covers a fresh installation, an in-place upgrade, verification and rollback of DBDOME on a Windows Server host. It documents the offline (air-gapped) installation path, which is the supported default: **no internet access is required at any point**.

Read Section 3 (Prerequisites) and Section 5 (Pre-install checklist) in full before running any installer. Most failed installations are traced to a prerequisite in Section 3 rather than to the installer itself.

2. What DBDOME installs

Component	Windows service	Port	Purpose
DBDOME collection engine	DBDOME_dbanalytics	—	Collects metrics, evaluates rules, raises alerts
DBDOME web interface	DBDOME_web	8080/HTTPS	Configuration UI, APIs
Grafana OSS 12.2.0	DBDOME_Grafana	3000	Dashboards, alert views
Ollama runtime	DBDOME_ollama	11434 (loopback)	Serves the local AI model
PostgreSQL 18	postgresql-x64-18	5432	DBDOME repository database

Standard paths

C:\ProgramData\DBDOME\bin	program files, .env, legal notices
C:\ProgramData\DBDOME\data	Grafana data, dashboards, plugins
C:\ProgramData\DBDOME\ollama\models	AI model store (machine-wide)
C:\ProgramData\DBDOME\bin\logs	service logs

3. Prerequisites

3.1 Hardware

Resource	Minimum	Recommended
CPU	4 cores	8 cores
RAM	16 GB	32 GB
Free disk	60 GB	120 GB

The AI security agent holds roughly **3.6 GB resident** while the model is loaded. Size RAM accordingly, or leave the agent disabled (Section 8).

Disk sizing must account for the repository database growing with the number of monitored servers and the retention period. Alert and metric tables are partitioned, but partitions still consume space.

3.2 Operating system

- Windows Server 2019, 2022 or 2025 (x64). Windows 10/11 x64 is supported for evaluation only.

- An account with **local Administrator** rights.
- PowerShell 5.1 or later.

Elevation is mandatory. Service registration, firewall rules, `setx /M` and `icacls` all fail silently or partially from a non-elevated shell. Right-click the installer and choose **Run as administrator**.

3.3 Network and ports

Port	Direction	Purpose
8080/TCP	inbound	DBDOME web UI (HTTPS)
3000/TCP	inbound	Grafana
5432/TCP	local	PostgreSQL
11434/TCP	loopback only	Ollama. Must not be exposed.
target DB ports	outbound	1433 MSSQL, 1521 Oracle, 3306 MySQL/MariaDB, 5432 PostgreSQL, 27017 MongoDB, 8123 ClickHouse
25/587/465	outbound	SMTP for alert mail, if used
514	outbound	Syslog/SIEM, if used

3.4 Accounts required

- **Local Administrator** on the DBDOME host.
- A **read-only monitoring login on each target database**. DBDOME requires only

the privileges needed to read system and dynamic management views. Do not grant `sysadmin`, `DBA` or ownership. Blocking and masking features, if enabled later, require additional privileges — grant those only when the feature is deliberately adopted.

3.5 Legal prerequisite

Read `EULA.txt` and `THIRD_PARTY_NOTICES.txt` at the root of the installation media **before installing**. Installation constitutes acceptance of the DBDOME EULA and of the Microsoft license terms governing the bundled ODBC driver.

Confirm you have a lawful basis for monitoring before you point DBDOME at a production database — DBDOME observes statement text, login identities and client program names. In many jurisdictions this requires prior notification of employees, and may require works-council consultation or a data protection impact assessment. This is the customer's responsibility under EULA Section 4.1, and it must be settled before, not after, go-live.

4. Installation media

Media	Size	Use
<code>dbdome_setup</code>	~6.5 GB	Fresh installation
<code>dbdome_update</code>	~6.5 GB	Upgrade of an existing installation

Both are fully self-contained: PostgreSQL, the Microsoft ODBC driver, Grafana, the Ollama installer, the AI model in both runtime formats, and all SQL scripts.

Verify the media before use. Confirm the total size is within ~50 MB of the figure above. A short copy — common when media is transferred over an unreliable link — typically truncates the largest files, which are the AI model and the Ollama installer, and surfaces much later as an unexplained agent failure.

5. Pre-install checklist

Complete every line before starting.

- Installer will be run **as Administrator**
- Host meets Section 3.1 sizing
- Ports in Section 3.3 are free (`netstat -ano | findstr ":8080 :3000 :5432"`)
- No existing PostgreSQL on 5432 that must be preserved
- Read-only monitoring credentials obtained for each target
- Target databases reachable from this host (test with `Test-NetConnection`)

- Host has a **static LAN IP**, or a permanent DHCP reservation
- Full backup / VM snapshot taken (upgrades especially)
- EULA and third-party notices reviewed
- Lawful basis for monitoring confirmed
- Maintenance window agreed

Static IP matters. DBDOME records the host's LAN IP in `.env` as `ORG_IP` and in the database as `config.local_ip`. Grafana datasources and ingest URLs are built from it. If the address changes, dashboards and ingest break until it is refreshed. The updater refreshes it automatically (update step 13); a fresh install does not, so get it right at install time.

6. Fresh installation

6.1 Procedure

1. Copy the `dbdome_setup` media to a local drive. ****Do not run it from a network share or mounted ISO**** — several steps write back into the payload directory.
2. Right-click `dbdome_setup.exe` → **Run as administrator**.
3. Accept the EULA when prompted.
4. Provide the installation path (default `C:\ProgramData\DBDOME`) and the PostgreSQL superuser password when asked. **Record that password**; it cannot be recovered from the installation.
5. Let the installer run to completion without interruption.

6.2 What the installer does

The installer prints numbered steps to the console and to its log. Note that the **printed step numbers do not run in numeric order** — steps 16 and 17 were added later and execute out of sequence. This is cosmetic; follow the console order, not the numbers.

Order	Step	Action
1	1	Copy DBDOME files
2	2	Install Microsoft ODBC Driver for SQL Server
3	3	Install PostgreSQL 18.3
4	4	Configure <code>pg_hba.conf</code>
5	5	Create database and users
6	6	Restore the <code>dbanalytics</code> baseline database
7	7	Configure Grafana
8	8	Create configuration files (<code>.env</code>)
9	17	Provision the security-agent model (Ollama + model store)
10	9	Configure Windows Firewall
11	16	Trust the DBDOME code-signing certificate
12	10	Install DBDOME <code>dbanalytics</code> Windows service(s)
13	11	Start DBDOME services
14	12	Validate installation
15	13	Create desktop shortcut
16	14	Install the DBDOME Grafana Windows service

Order	Step	Action
17	15	Restart Grafana to clear plugin issues

6.3 Expected duration

25–50 minutes, dominated by the PostgreSQL install, the database restore and the Ollama/model provisioning step.

7. Post-install verification

Run every check. An installer that reported success is not proof that the platform is collecting.

7.1 Services

```
Get-Service DBDOME_dbanalytics, DBDOME_web, DBDOME_Grafana, DBDOME_Ollama, postgresql-x64-18 |
Format-Table Name, Status, StartType
```

All five must be **Running** and **Automatic**.

7.2 Web interfaces

- <https://<host>:8080> — DBDOME configuration UI
- <http://<host>:3000> — Grafana

If 8080 is unreachable from another machine but works locally, the cause is almost always the Windows Firewall profile. When the network adapter is classified as **Public**, the DBDOME rule may be scoped to Private/Domain only and the port is silently dropped. Check with: `powershell Get-NetConnectionProfile Get-NetFirewallRule -DisplayName "DBDOME" | Get-NetFirewallProfile` Either reclassify the adapter as Private, or widen the rule's profile.

7.3 Database

```
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U postgres -d dbanalytics -c "\dn"
```

Expect schemas including `monitoring`, `alerts`, `metrics`, `config`, `rootcause`, `log`.

7.4 Collection is actually running

This is the check that matters. Roughly 10 minutes after start-up:

```
SELECT max(entry_date) FROM monitoring.general_metric_metadata_results;
```

The timestamp must be recent and must keep advancing. If it is static, no collection is occurring regardless of service state.

7.5 Local IP recorded correctly

```
Select-String -Path "C:\ProgramData\DBDOME\bin\.env" -Pattern "^ORG_IP="
```

It must equal the host's LAN IP, not `127.0.0.1` and not a stale address.

8. Security agent (AI triage) — optional

The agent reads the query that tripped a rule, compares it against what that server has been observed running, and writes a plain-language reason onto the alert. It runs **after** the alert is dispatched, so it never delays alerting.

8.1 Enable

In `C:\ProgramData\DBDOME\bin\.env`:

```
SECURITY_AGENT_ENABLED=true
```

Restart `DBDOME_dbanalytics`. That is the entire minimum configuration — everything else has a working default.

8.2 Required database objects

Apply once, in this order. All are idempotent.

Script	Status
<code>7600_security_agent_verdict.sql</code>	Required — verdict audit trail
<code>7620_gmmr_server_entry_index.sql</code>	Strongly recommended — see below
<code>7630_security_agent_results_et.sql</code>	Required — reads verdicts back
<code>7640_security_agent_annotation_process.sql</code>	Required — registers the sweep; needs 7630
<code>7610_ransomware_sequence.sql</code>	Optional — ransomware sequence rule

7620 is close to mandatory in practice. Without it the precedent lookup was measured at 3.19 s warm and roughly 50 minutes cold, because the query plan walked 678,529 rows of a partition to find none for that server. With the index: 0.023 s. It takes a `SHARE` lock while building — apply it during a collection pause.

7610 installs INACTIVE by design. Review `ransomware_min_distinct_ops` (4) and `ransomware_window_minutes` (5) in `config.global_params` against the site's real traffic before enabling, or routine nightly maintenance will alert every night.

8.3 Verify the agent

```
cd C:\ProgramData\DBDOME\bin
.\dbdome_dbanalytics.exe --check-security-agent
```

Reports the resolved backend and model availability. `available: True` means it will classify. `False` means every alert **fails open** — alerts still fire, they simply arrive without a reason attached. That is safe, but the agent contributes nothing.

After the first alerts:

```
SELECT decided_by, count(*), round(avg(elapsed_ms))
FROM alerts.security_agent_verdict GROUP BY 1;
```

`decided_by='model'` is a genuine verdict. Any other value means the agent failed open, and the value states why.

8.4 Confirm Ollama is owned by the service

The Ollama vendor installer is **per-user** and adds a tray shortcut to the installing account's Startup folder. That tray app binds port 11434 as the logged-in user, while DBDOME runs as `LocalSystem`. When that user is not logged in, the port is unowned and the agent falls back to the slower sidecar.

Checking the service state is **not sufficient** — it will report Running and `/api/tags` will answer while the port belongs to the interactive user. Check port ownership instead:

```
Get-NetTCPConnection -LocalPort 11434 -State Listen |
ForEach-Object { (Get-Process -Id $_.OwningProcess).ProcessName }
```

The owning process must run as **SYSTEM**. If not, disable the `ollama.lnk` Startup shortcut, end `ollama app.exe`, then restart `DBDOME_ollama`.

8.5 Operating limits — read before relying on it

- Verdicts are **not yet trustworthy enough to suppress alerts**. Benign queries

have been misclassified, and reasons have contained factual slips about the query. Annotate-only is the supported mode; suppression is off by default and should stay off.

- The agent needs precedent. With no recent rows in

`monitoring.general_metric_metadata_results` it has nothing to compare against and will flag everything — correct by its own logic, useless in practice. Collection must be healthy first.

- Triage takes 20–60 s per alert depending on backend. This never delays the alert itself.

9. Upgrading an existing installation

9.1 Before you start

- **Full database backup**, verified restorable
- VM snapshot if available
- Current version and `.env` recorded
- Maintenance window agreed — collection stops during the upgrade

9.2 Procedure

1. Copy the `dbdome_update` media locally.
2. Right-click `dbdome_update.exe` → **Run as administrator**.
3. Let it run to completion.

The updater performs, in printed order:

Step	Action
1	Preflight checks
2	Stop DBDOME services
3	Replace files, reinstall services
4	Start DBDOME services
5	Stop Grafana
6	Replace <code>grafana.db</code>
7	Start Grafana
8	Base schema install
9	Run numbered SQL scripts
10	Encrypt secrets (<code>.env</code> and stored passwords)
11	Trust code-signing certificate
13	Refresh local IP in <code>.env</code> and <code>config.local_ip</code>
12	Provision security-agent model

As with setup, printed step numbers are not in execution order.

9.3 Preserving local customization

`.env` is not fully preserved by a file replace. The payload ships a template `.env`. The updater merges managed keys, but any key you added manually should be recorded before upgrading and re-checked afterwards.

Back up before upgrading:

```
Copy-Item C:\ProgramData\DBDOME\bin\.env `
  "C:\ProgramData\DBDOME\bin\.env.backup_$(Get-Date -f yyyyMMdd)"
```

`rules.json` under `bin\_internal\security_agent\` is editable on the installed box and **is replaced by an upgrade**. Back it up if you have customized it.

9.4 After upgrading

Repeat every check in Section 7, plus:

```
SELECT max(entry_date) FROM monitoring.general_metric_metadata_results;
```

Confirm collection resumed. Then confirm the scheduler picked up all processes:

```
SELECT process_name, is_active FROM metrics.registered_processes ORDER BY 1;
```

A process row can be `is_active=true` and still never run, if the build does not map that name to a function. The symptom is silence rather than an error. If a feature is enabled but producing nothing, check the service log for `Unknown process name` entries.

10. Rollback

1. Stop all DBDOME services.
2. Restore the VM snapshot, **or**: restore the database backup and restore the previous `bin` directory and `.env`.

3. Start services and verify with Section 7.

There is no automated downgrade. Rollback depends entirely on the backup taken in Section 9.1 — this is why that step is not optional.

11. Troubleshooting

Symptom	Likely cause	Action
Services will not start	Non-elevated install, or PostgreSQL down	Check <code>bin\logs</code> ; confirm <code>postgresql-x64-18</code> is running
8080 works locally, not remotely	Firewall profile is Public	Section 7.2
Grafana loads, panels empty	Datasource points at a stale IP	Check <code>ORG_IP</code> ; re-run updater to refresh
No alerts at all	Collection not running	Section 7.4
Alerts fire with no reason attached	Agent disabled, no model, or sweep not registered	Sections 8.1–8.3
Agent slow, frequent timeouts	Ollama port owned by interactive user	Section 8.4
Login failed for <user> (18456) plus 4060 Cannot open database None	Target configured with a NULL database, not bad credentials	Set the database on the target definition; connections default to master
Installer fails on file copy, access denied	Payload ACLs restrict the running account	Re-run elevated; if it persists, grant modify rights on the payload directory
SQL script reports errors but exit 0	Script partially applied	Read the end-of-run summary; a non-zero error-line count is a failure even at exit 0

Log locations

<code>C:\ProgramData\DBDOME\bin\logs\</code>	service logs
<code>%TEMP%\dbdome_setup*.log</code>	installer log
<code>%TEMP%\dbdome_update*.log</code>	updater log

Also check the `log` schema in the database, which records operational errors the services could not write to disk.

12. Appendix A — `.env` reference

`C:\ProgramData\DBDOME\bin\.env`. Changes require a `DBDOME_dbanalytics` restart.

Core

Key	Purpose
PG_HOST, PG_PORT, PG_DB, PG_USER, PG_PASSWORD	Repository database connection
ORG_IP	Host LAN IP. Must be the real address, never loopback
DBDOME_SECRET_KEY	Encryption key for stored credentials
GRAFANA_EXE, GRAFANA_WATCHDOG_INTERVAL	Grafana process management
ORACLE_CLIENT_LIB_DIR	Oracle client path, if monitoring Oracle
LLM	Language-model integration setting

PG_PASSWORD and stored target passwords are encrypted at rest with the enc:v1: prefix. DBDOME_SECRET_KEY is **required to decrypt them**. Back it up somewhere other than the DBDOME host. Losing it means re-entering every stored credential.

Security agent

Key	Default	Notes
SECURITY_AGENT_ENABLED	false	The only key you must set
SECURITY_AGENT_BACKEND	auto	auto / ollama / sidecar
SECURITY_AGENT_SUPPRESS	false	Leave off. Lets a verdict withhold an alert
SECURITY_AGENT_ANNOTATE_INLINE	false	Leave off. Puts triage back in the alert path
SECURITY_AGENT_ANNOTATE_BATCH	25	Alerts annotated per sweep
SECURITY_AGENT_ANNOTATE_LOOKBACK_MIN	120	How far back the sweep looks
SECURITY_AGENT_OLLAMA_MODEL	phi3:mini-128k	
SECURITY_AGENT_OLLAMA_URL	http://127.0.0.1:11434	Loopback; nothing leaves the host
SECURITY_AGENT_OLLAMA_KEEP_ALIVE	24h	-1 keeps the model loaded indefinitely
SECURITY_AGENT_TIMEOUT_SECS	90	Per alert; exceeded means no reason attached
SECURITY_AGENT_MODEL	bundled GGUF	Sidecar backend only
SECURITY_AGENT_DOMAINS	*	Which alert domains are annotated
SECURITY_AGENT_MIN_CONFIDENCE	0.75	Consulted only if suppression is on

13. Appendix B — Licensing

DBDOME is proprietary software licensed under EULA.txt. It is distributed with third-party components under their own licenses, documented in THIRD_PARTY_NOTICES.txt with full texts in bin\legal\licenses\.

Points that affect installation decisions:

- **Microsoft ODBC Driver 18** is proprietary. Microsoft's own terms apply directly to you and are supplied verbatim on the media.
- **Grafana 12.2.0 and its plugins** are AGPL-3.0. They are redistributed

unmodified. Your rights under the AGPL are not limited by the DBDOME EULA, and corresponding source is available on written offer.

- **Ollama and the Microsoft Phi-3 model** are MIT licensed.
- **PostgreSQL** is under the PostgreSQL License.
- **psycopg2 and paramiko** are LGPL. You may replace them with your own build.

Nothing in the DBDOME EULA restricts any right granted by an open source license.

14. Support

Technical support	support@dbexpert.ai
Licensing	legal@dbexpert.ai
Open source requests	opensource@dbexpert.ai

When raising an installation issue, attach the installer log from %TEMP%, the output of the Section 7 verification commands, and the DBDOME version.

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